

Black Litterman vs Naive Mean-Variance Optimization

MF825: Portfolio Construction Final Project - Team 7

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Executive Summary

Research Question: Does Black-Litterman improve portfolio construction relative to Naive Mean-Variance?

- We use **49 Fama-French industry portfolios** and a rolling daily backtest
- Naive MV uses **historical sample means** directly
- **Black-Litterman combines:**
 - Market-implied equilibrium returns
 - Investor views based on trailing industry spread
- **Main finding:**
 - Naive MV has higher raw return and Sharpe
 - BL has lower volatility, lower turnover, and much lower concentration
- **Conclusion:**
 - BL is more stable and interpretable, but our chosen trailing views did not outperform Naive MV on return or Sharpe ratio

Black-Litterman Framework

Black-Litterman is a portfolio optimization framework that replaces noisy sample mean returns with “posterior” expected returns

- It starts from a benchmark portfolio and asks:
 - What expected returns are implied by the market portfolio?
- These implied returns are treated as the equilibrium prior
- Investor views are then added as relative return forecasts
- The model combines:
 - the equilibrium prior μ_E
 - the view matrix P
 - the view returns Q
 - uncertainty from the covariance matrix V
- The output is a new expected-return vector

Why Compare Black-Litterman to Naive Mean-Variance?

- Naive MV uses historical sample means directly as expected returns.
- Sample means are noisy, especially with **49 industry portfolios**.
- Small changes in expected returns can produce extreme portfolio weights.
- Black-Litterman starts from market-implied returns and blends in views.
- Our test: compare return, Sharpe, turnover, concentration, and rolling performance.

Methodology

Data and Benchmark Construction

- **Data source:** Fama-French 49 Industry Portfolios, daily returns
- **Factor data:** Daily Mkt-RF and RF from Fama-French Research Factors
- Industry excess returns are computed as:
$$R_{i,t}^{excess} = R_{i,t} - RF_t$$
- Start the sample at 1975 with complete industry and factor data
- **Benchmark weights** come from monthly Average Firm Size section
- Monthly firm-size weights are normalized to sum to 1 and mapped to daily dates

Benchmark portfolio

$w_{M,t}$ = normalized industry firm-size weights

- Monthly weights carried forward to daily dates.
- If size weights are missing, previous weights are carried forward.

Black-Litterman Implementation: Expected Returns

Calculated at each date t , using the previous 252 trading days

$$\mu_{BL,t} = [(\tau V_t)^{-1} + P^T D^{-1} P]^{-1} [(\tau V_t)^{-1} \mu_{E,t} + P^T D^{-1} Q_t]$$

1. **Estimate annualized covariance:**

$$V_t = \text{Cov}(R_{t-251:t}^{\text{excess}}) \times dpy$$

2. **Estimate rolling market risk aversion:**

$$g_t = \frac{\bar{R}_{M,t}}{\text{Var}(R_{M,t})}$$

3. **Compute equilibrium returns:**

$$\mu_{E,t} = g_t V_t w_{M,t}$$

4. **Define P , test 3 specifications for Q : trailing, momentum, and mean reversion**

$$P = \begin{bmatrix} 0 & \cdots & +1_{\text{Softw}} & \cdots & -1_{\text{Telcm}} & \cdots & 0 \\ 0 & \cdots & +1_{\text{Chips}} & \cdots & -1_{\text{Util}} & \cdots & 0 \\ 0 & \cdots & +1_{\text{Hlth}} & \cdots & -1_{\text{Steel}} & \cdots & 0 \end{bmatrix}$$

5. **Estimate trailing relative views over 63 trading days:**

$$Q_t = \begin{bmatrix} \mu_{\text{Softw}} - \mu_{\text{Telcm}} \\ \mu_{\text{Chips}} - \mu_{\text{Util}} \\ \mu_{\text{Hlth}} - \mu_{\text{Steel}} \end{bmatrix}$$

6. **Combine equilibrium returns and views to get $\mu_{BL,t}$**

Key Assumptions

- Estimation window: 252 trading days
- View window: 63 trading days
- Prior uncertainty: $\tau = \frac{1}{252}$
- Ridge adjustment: $\lambda = 10^{-6}$
- View uncertainty: $D_t = \text{diag}(P V_t P^T)$

Portfolio Formation and Rolling Backtest

1. Estimate expected returns:

BL: $\mu_{BL,t}$

Naive MV: $\mu_{N,t}$

1. Solve the same long-only optimization problem:

$$\max_w \quad w' \mu - \frac{1}{2} \gamma w' V_t w$$

Subject to:

$$\sum_i w_i = 1, \quad w_i \geq 0$$

1. Optimizer risk aversion: $\gamma = 3$

2. Apply weights to next-day realized returns:

$$R_{p,t+1} = w_t' R_{t+1}$$

3. Roll forward one day and repeat

Optimization/Backtest Assumptions

- Long-only portfolios.
- Fully invested: weights sum to 1.
- No short selling.
- Same covariance matrix used for BL and Naive MV.
- Same optimizer used for both strategies.
- One-day holding period.
- Next-day returns used to avoid look-ahead bias.

Comparison Portfolios and Diagnostics

Main comparison:

- Naive MV: uses sample mean returns directly
- Black-Litterman: uses equilibrium returns plus trailing relative views

Diagnostic benchmarks:

- Views Only: isolates the effect of views without BL shrinkage
- Market Size: passive size-weighted industry benchmark

Performance Metrics

- Annualized return: $\bar{R}_p \times dpy$
- Annualized Volatility: $\sigma_p \times \sqrt{dpy}$
- Sharpe Ratio: $\frac{\text{Annualized Return}}{\text{Annualized Volatility}}$
- Turnover: Average $\left(\sum_i |w_{i,t} - w_{i,t-1}| \right)$
- Concentration: Average $\left(\sum_i w_{i,t}^2 \right)$

Results

Full Sample Performance Comparison

Strategy	Ann. Return	Ann. Vol	Sharpe	Turnover	Concentration
Black-Litterman	32.1%	26.5%	1.21	0.112	0.401
Naive MV	64.7%	37.0%	1.75	0.130	0.764
Views Only	62.5%	31.0%	2.02	0.144	0.964
Market Size	30.5%	21.8%	1.40	0.001	0.053

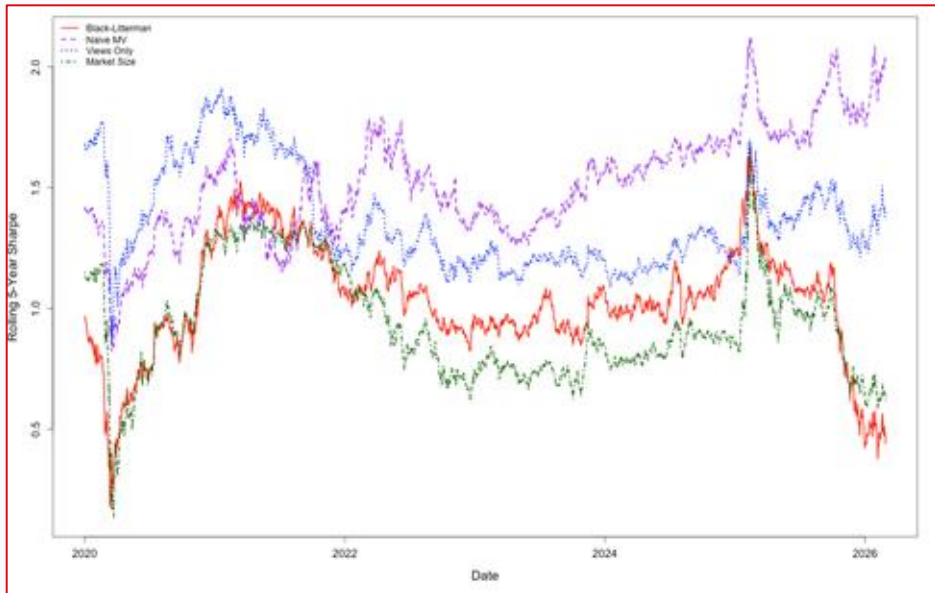
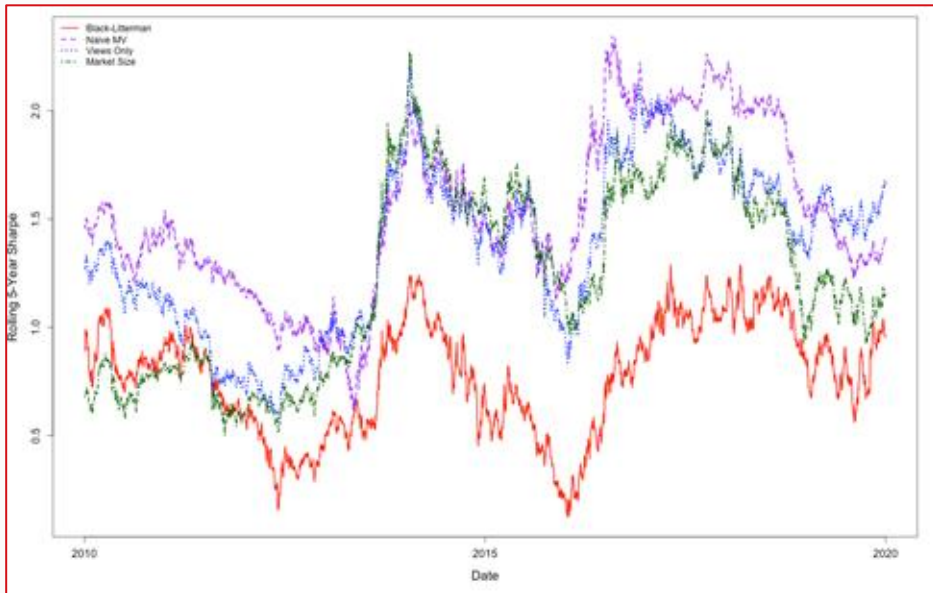
- Naive MV has the highest annual return
- Views Only has the highest Sharpe, but is extremely concentrated
- BL has lower return, but much lower volatility and concentration than Naive MV
- Market Size is the lowest-risk and lowest-turnover benchmark
- Different trailing views had marginal impact on BL performance

Did the Better Strategy Depend on the Time Period?

Rolling 5-Year Sharpe Ratios

2010s

2020s

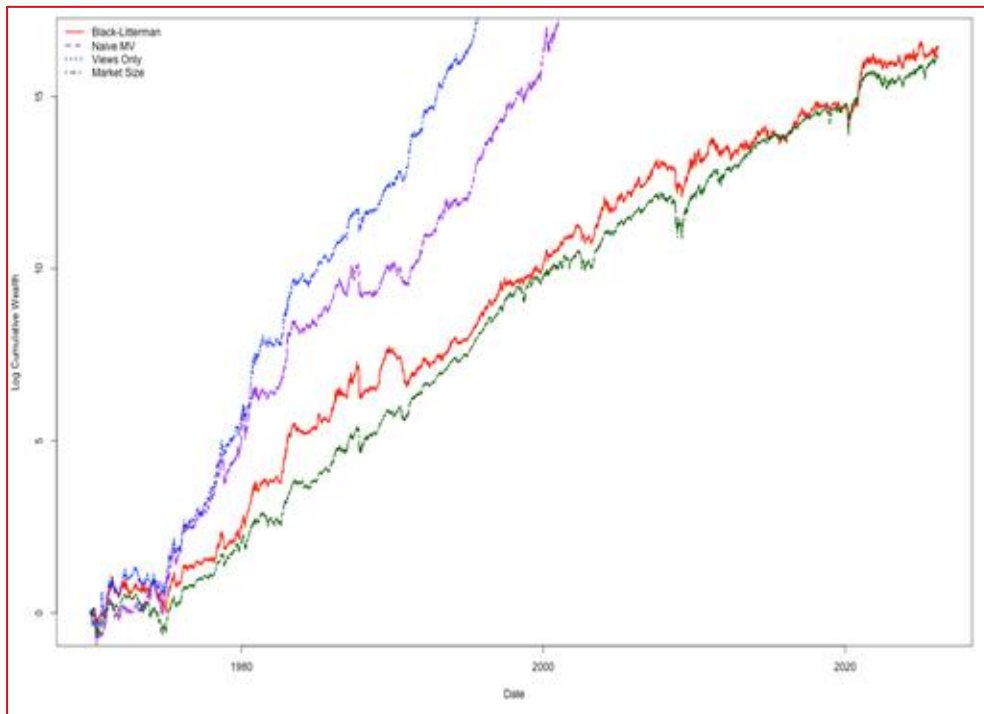


- MV often wins on rolling Sharpe, but BL gives a more stable and less concentrated allocation across regimes.

Cumulative Performance: BL is Smoother, Naive MV is More Aggressive

- Naive MV compounds faster over the full sample
- BL tracks the broad upward trend but with less extreme behavior
- Views Only grows explosively early, showing the danger of unconstrained views
- Market Size is the most diversified benchmark
- BL sits between passive market size and aggressive Naive MV

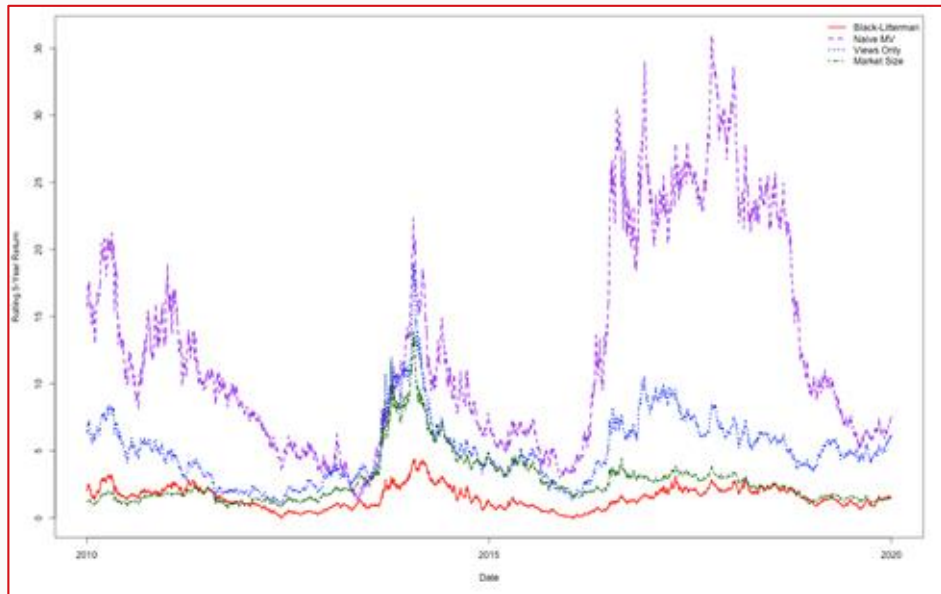
Log Cumulative Wealth Performance



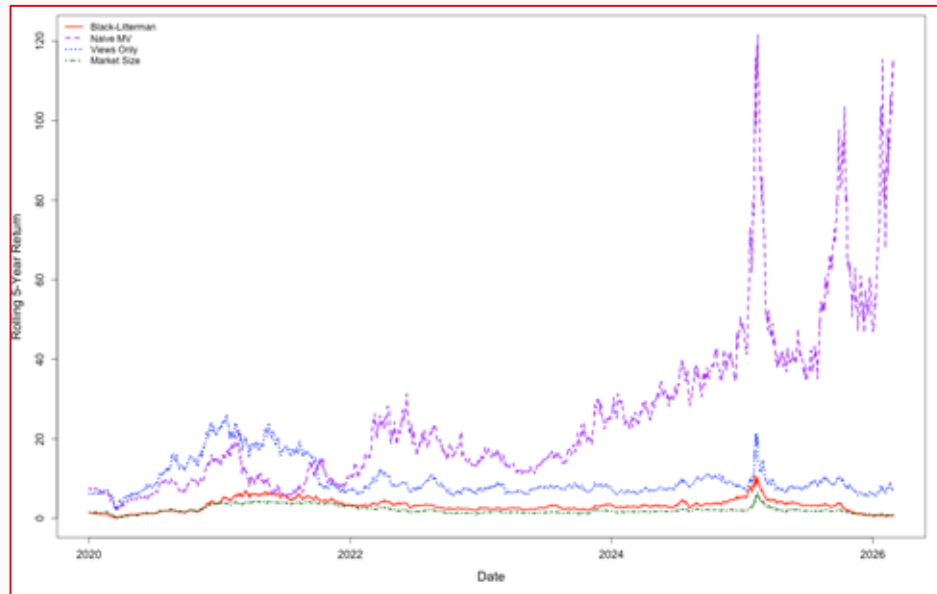
Rolling Window Performance

Rolling 5-Year Returns

2010s



2020s



Risk Diagnostics: BL Reduces Volatility & Concentration

- Naive MV often places very large weight in one industry
- Views Only is almost always dominated by one or a few industries
- Market Size is naturally diversified
- BL is much less concentrated than Naive MV and Views Only
- This is the clearest evidence that BL stabilizes the optimizer

Weight Stability: All Strategies, 2020s



Attribution and Trading Costs

CAPM Attribution

Strategy	Daily Alpha	Beta
BL	0.000267	0.709
Naive MV	0.000905	0.787
Market Size	0.000194	0.849

Effect of Adding a 10 BP Transaction Cost

Strategy	Absolute Change in Return	Change in Sharpe
BL	-5.5pp	-0.21
Naive MV	-6.4pp	-0.18
Market Size	0	0

- Naive MV still outperforms after transaction costs
- BL behaves more defensively and has lower beta than Naive MV and Market Size
- Transaction costs reduce Naive MV more in absolute return terms because it has higher turnover
- Proportional impact is larger for BL because BL's gross return is lower
- Market Size is barely affected because turnover is almost zero

Portfolio Composition: Top 5 Industries Allocations

Black-Litterman

Rank	Top Industries	Avg Weight
1	Utilities	14.5%
2	Smoke	11.5%
3	Beer	7.7%
4	Hardware	5.6%
5	Gold	4.8%

Mean-Variance

Rank	Top Industries	Avg Weight
1	Gold	12.6%
2	Coal	9.7%
3	Smoke	7.4%
4	Soda	5.6%
5	Health	4.4%

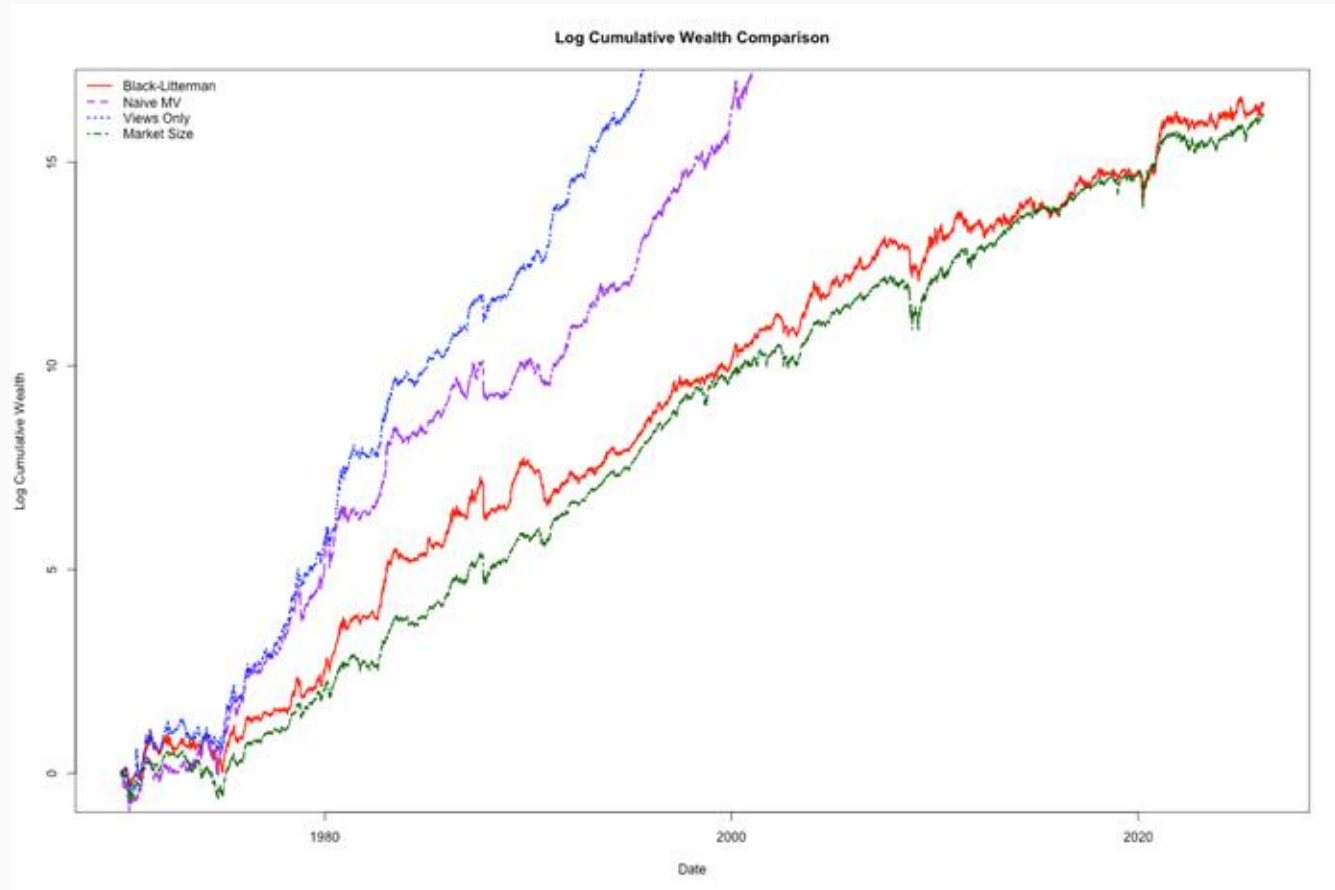
- BL has defensive and diversified exposures: Utilities, Smoke, Beer, Hardware
- Naive MV leans more into high-return sample winners: Gold, Coal, Smoke
- Naive MV allocations are more sensitive to historical mean estimates
- BL shrinks views toward equilibrium, reducing extreme allocations

Conclusion and Future Improvements

- We tested Black-Litterman against Naive MV using 49 industry portfolios
- Naive MV achieved higher return and Sharpe in the full out-side sample
- However, Naive MV was much more concentrated and volatile
- BL reduced: Volatility, beta, concentration, extreme dependence on sample means
- Main conclusion:
 - **BL is better as a disciplined portfolio construction framework**
 - **Naive MV performed better here, but with less stable allocations**
- Future work:
 - Test different relative view definitions
 - Tune view confidence
 - Add sector/industry weight caps
 - Evaluate shorter subperiods and alternative transaction costs

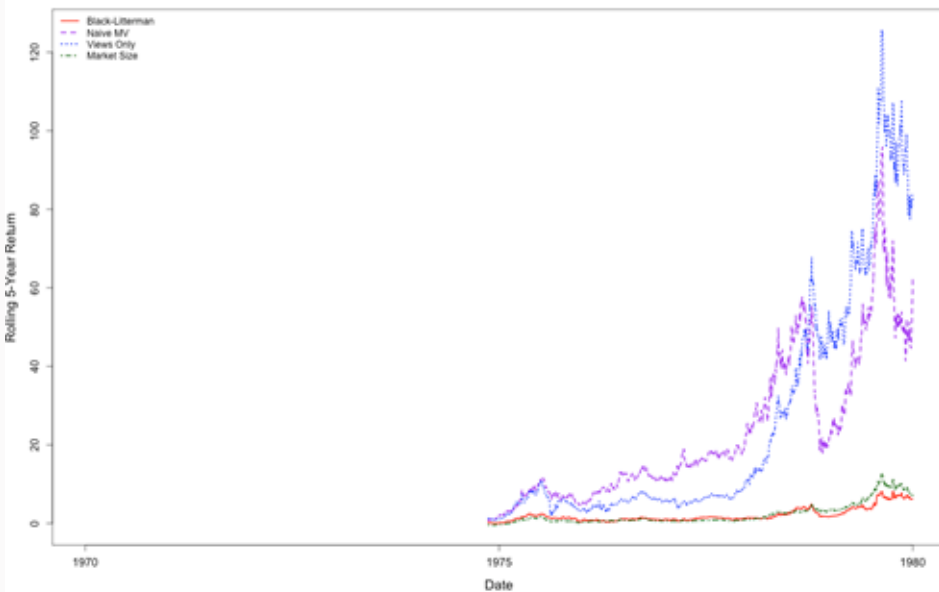
Appendix

Trailing Log Cumulative Wealth

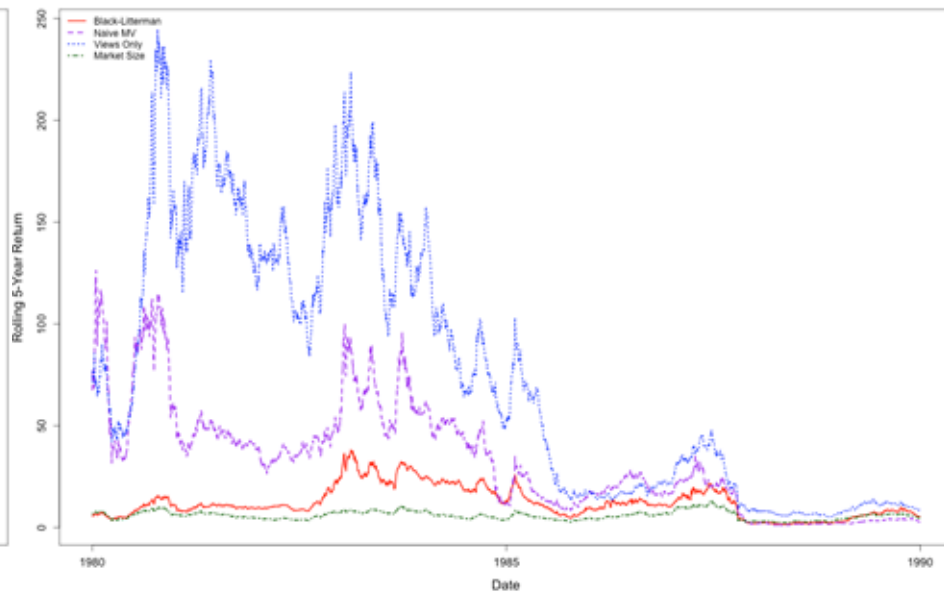


Rolling 5-Year Returns

Rolling 5-Year Return: 1970s

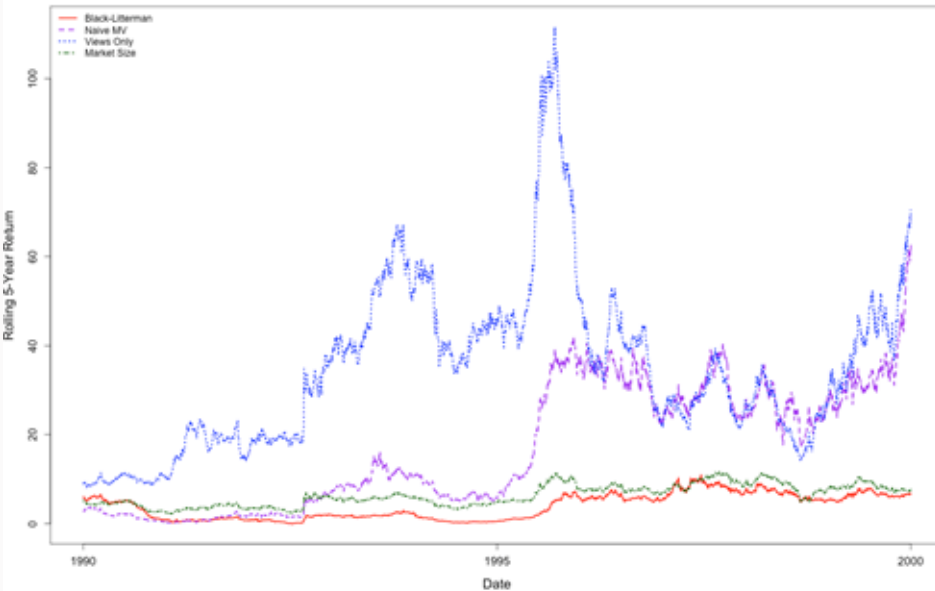


Rolling 5-Year Return: 1980s

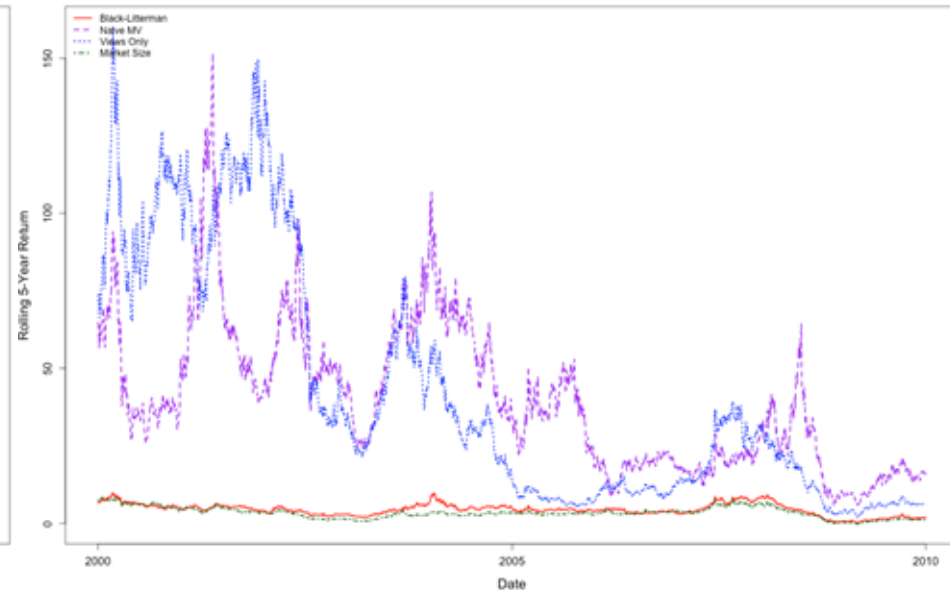


Rolling 5-Year Returns

Rolling 5-Year Return: 1990s

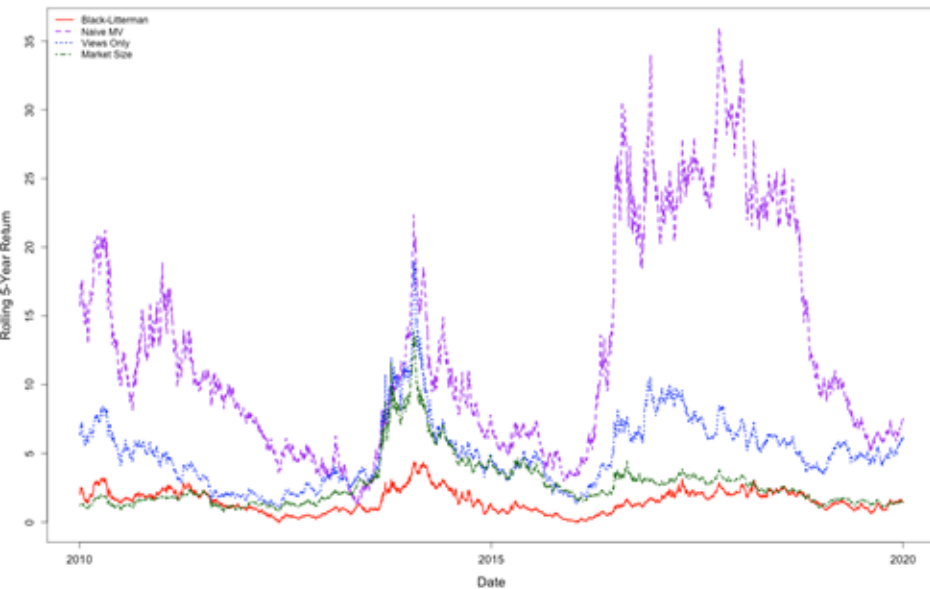


Rolling 5-Year Return: 2000s

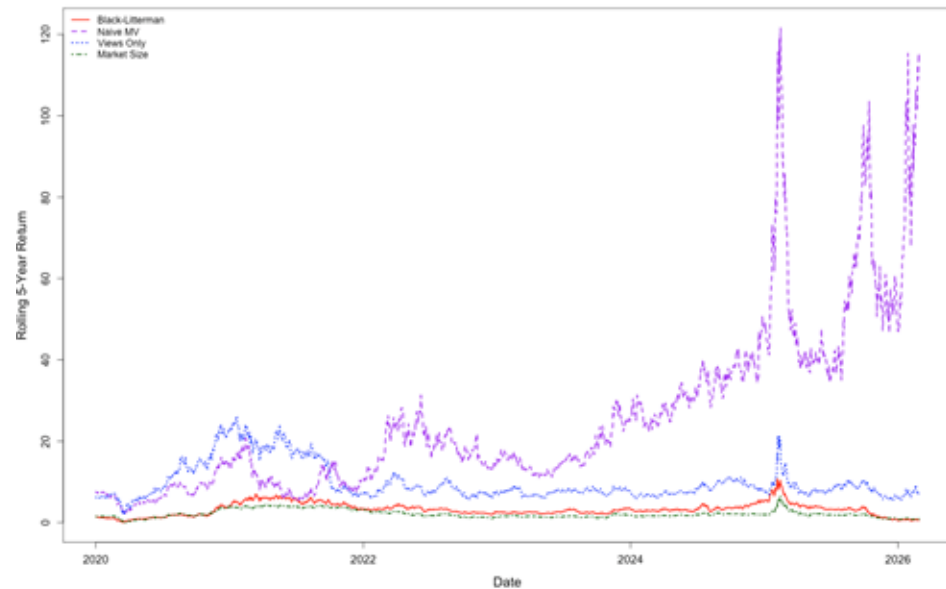


Rolling 5-Year Returns

Rolling 5-Year Return: 2010s

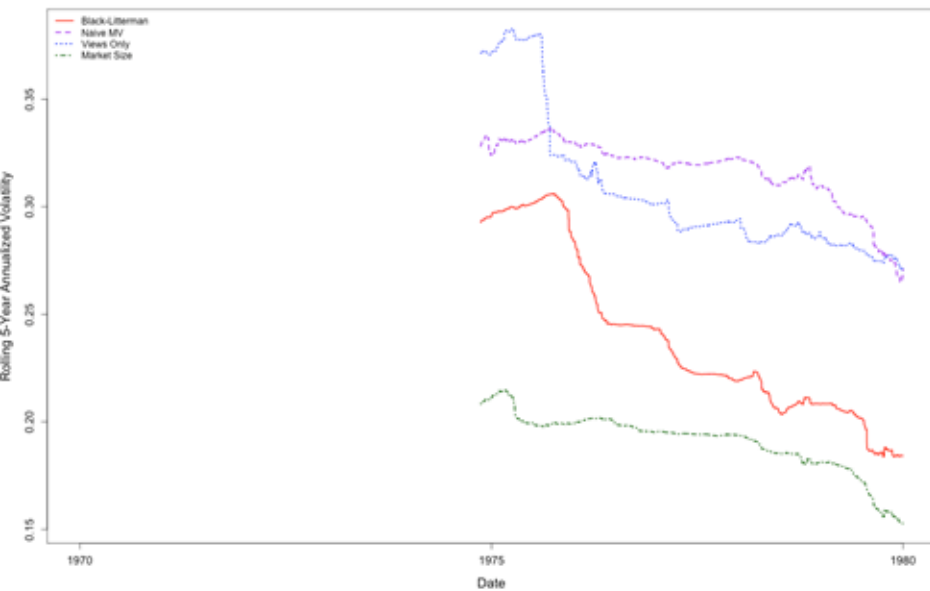


Rolling 5-Year Return: 2020s

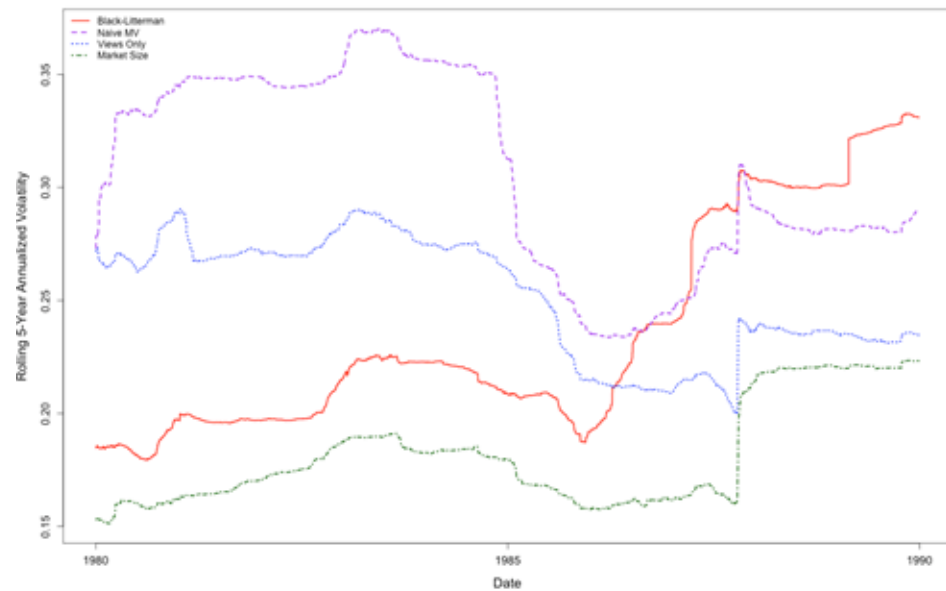


Rolling 5-Year Volatility

Rolling 5-Year Volatility: 1970s

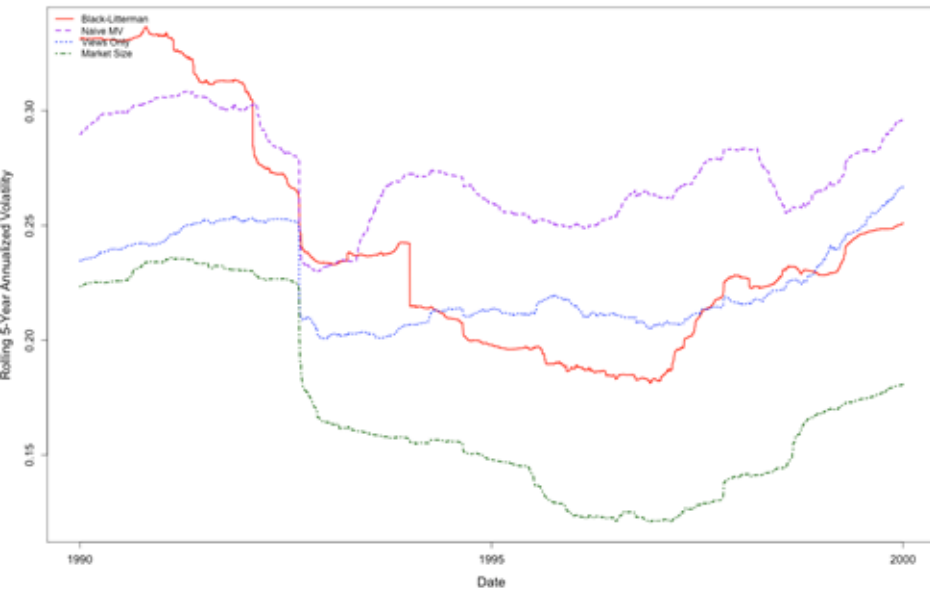


Rolling 5-Year Volatility: 1980s

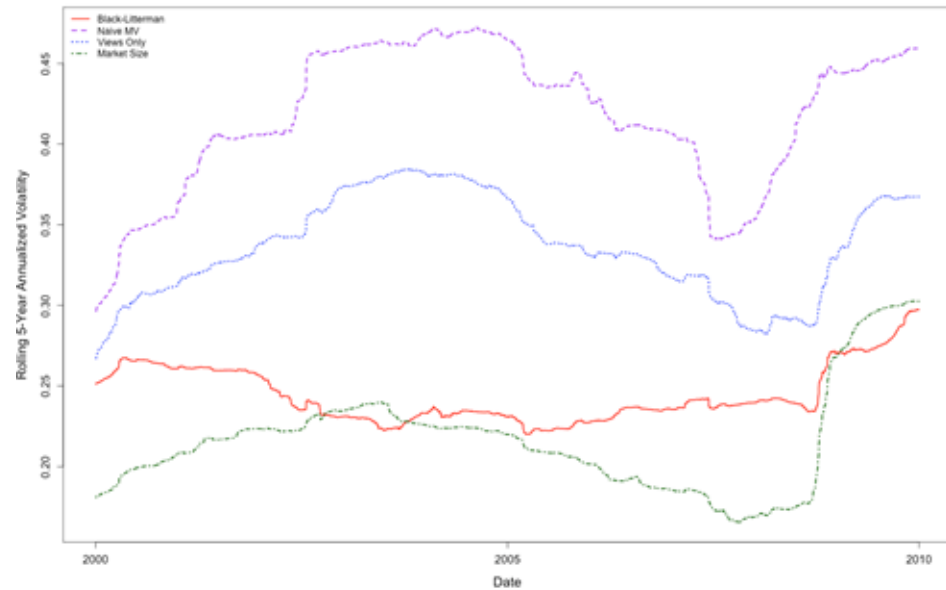


Rolling 5-Year Volatility

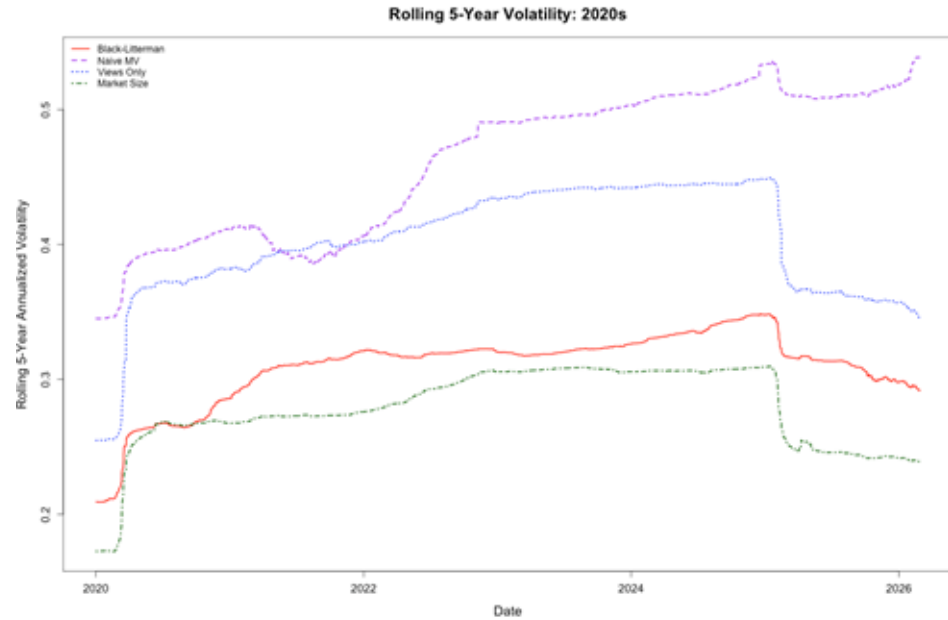
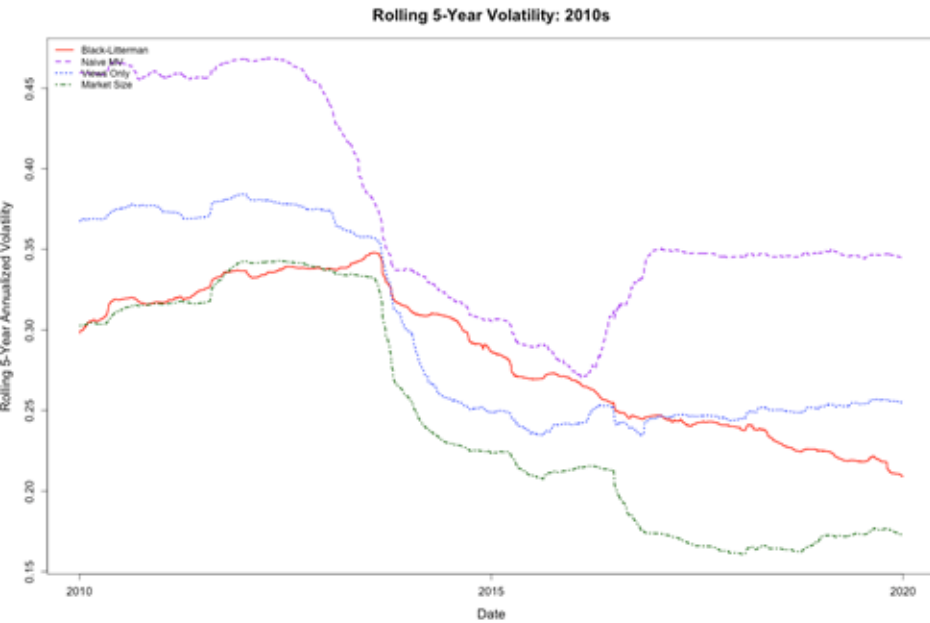
Rolling 5-Year Volatility: 1990s



Rolling 5-Year Volatility: 2000s

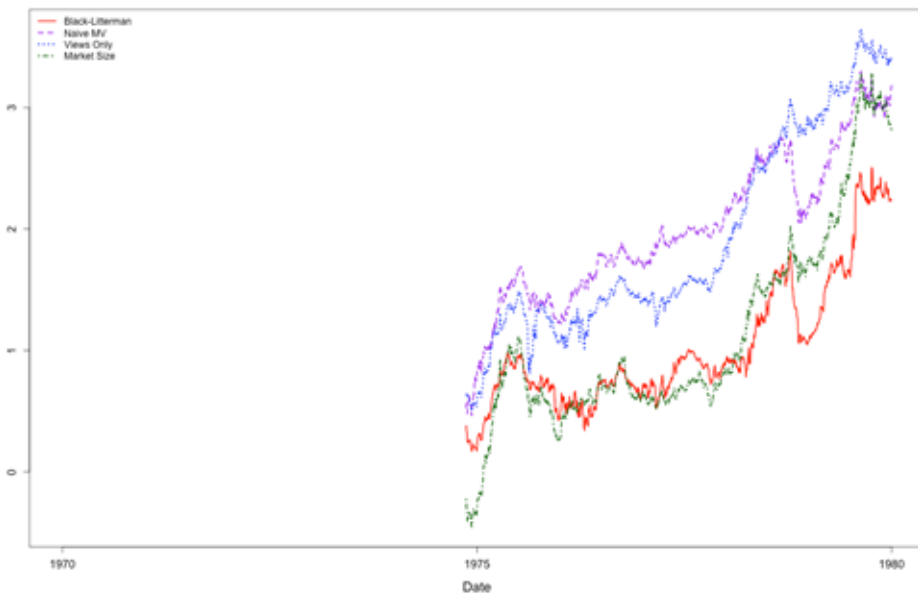


Rolling 5-Year Volatility

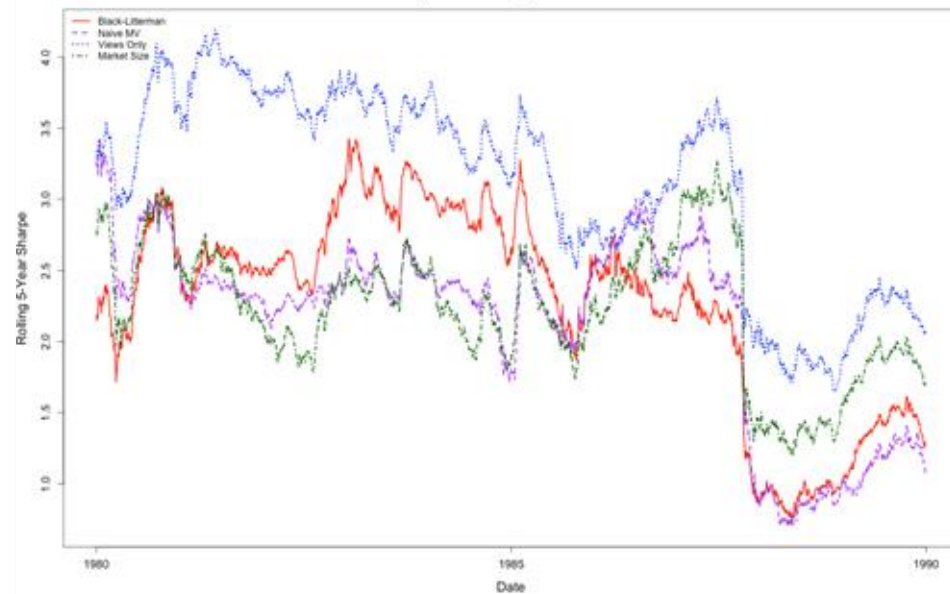


Rolling 5-Year Sharpe

Rolling 5-Year Sharpe Ratios: 1970s

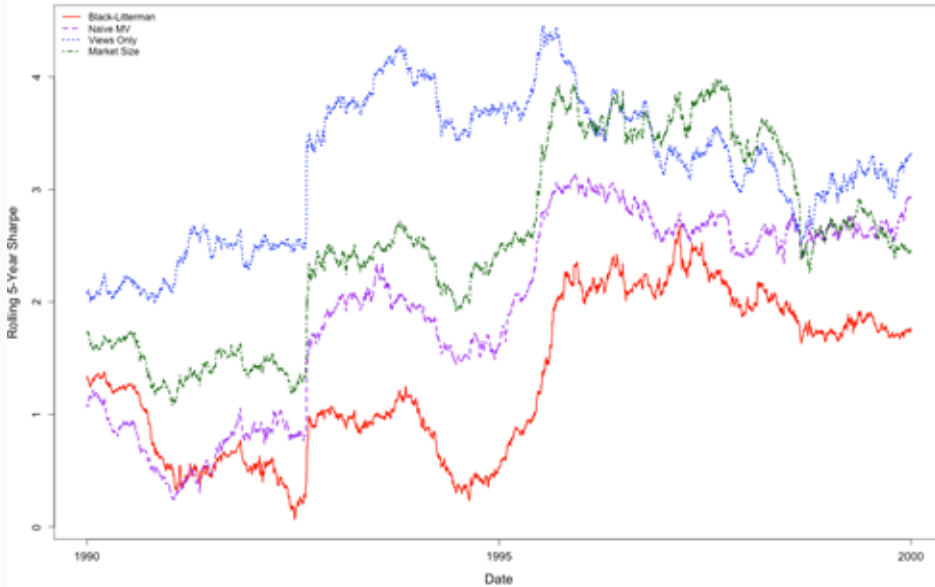


Rolling 5-Year Sharpe Ratios: 1980s

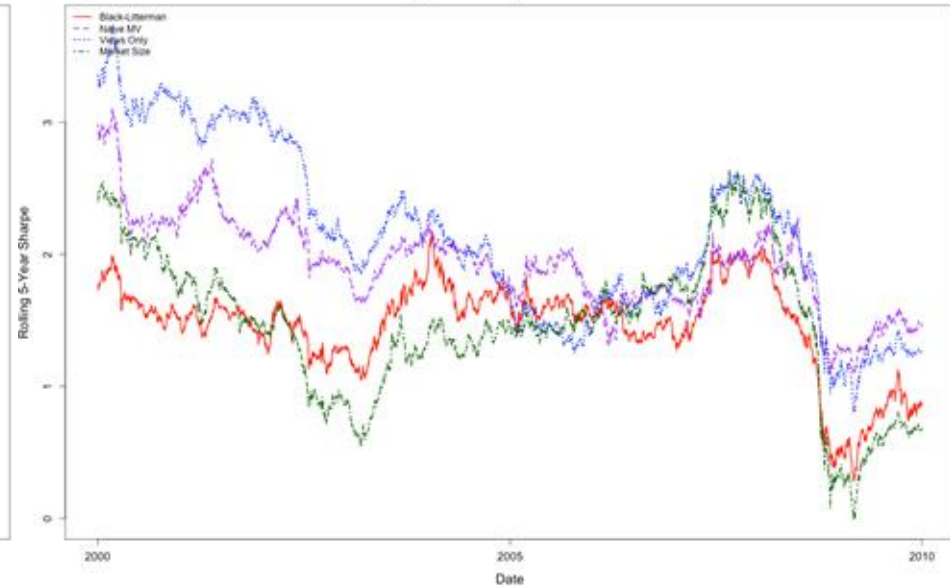


Rolling 5-Year Sharpe

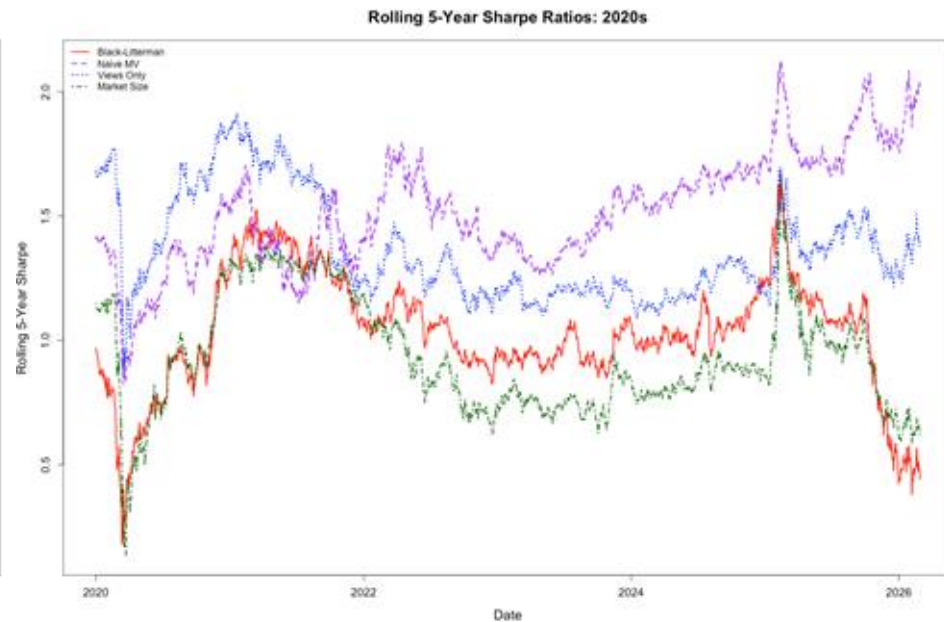
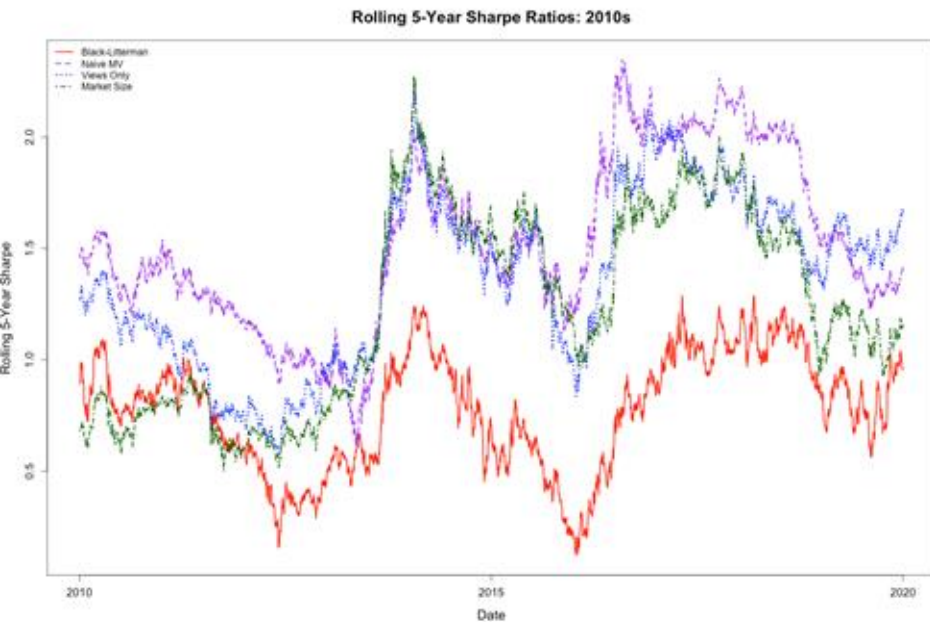
Rolling 5-Year Sharpe Ratios: 1990s



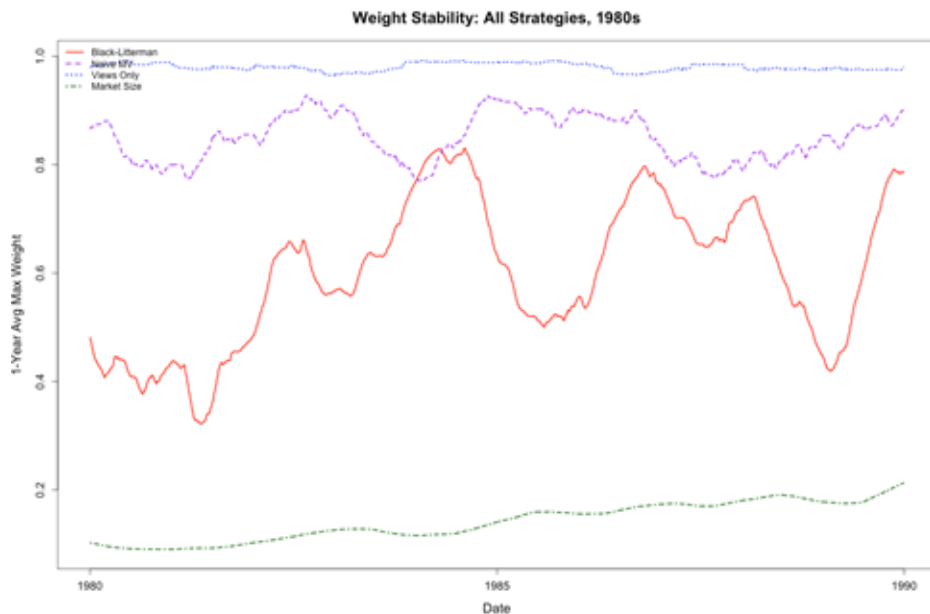
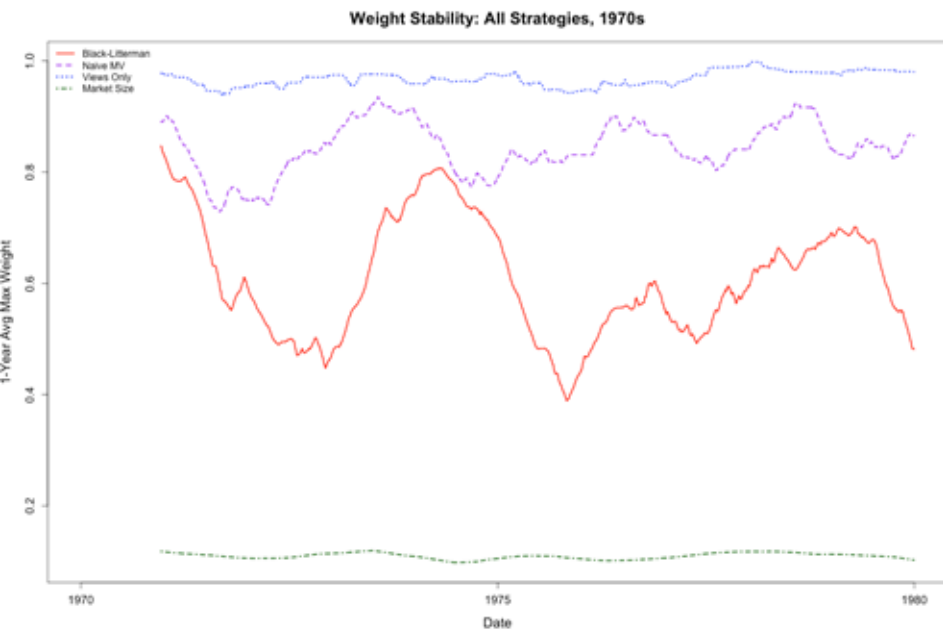
Rolling 5-Year Sharpe Ratios: 2000s



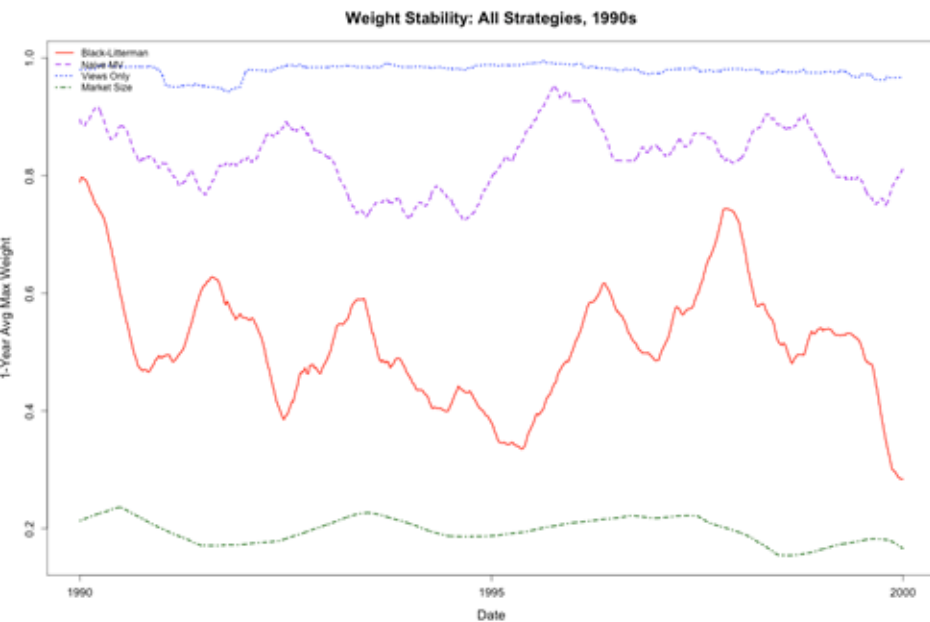
Rolling 5-Year Sharpe



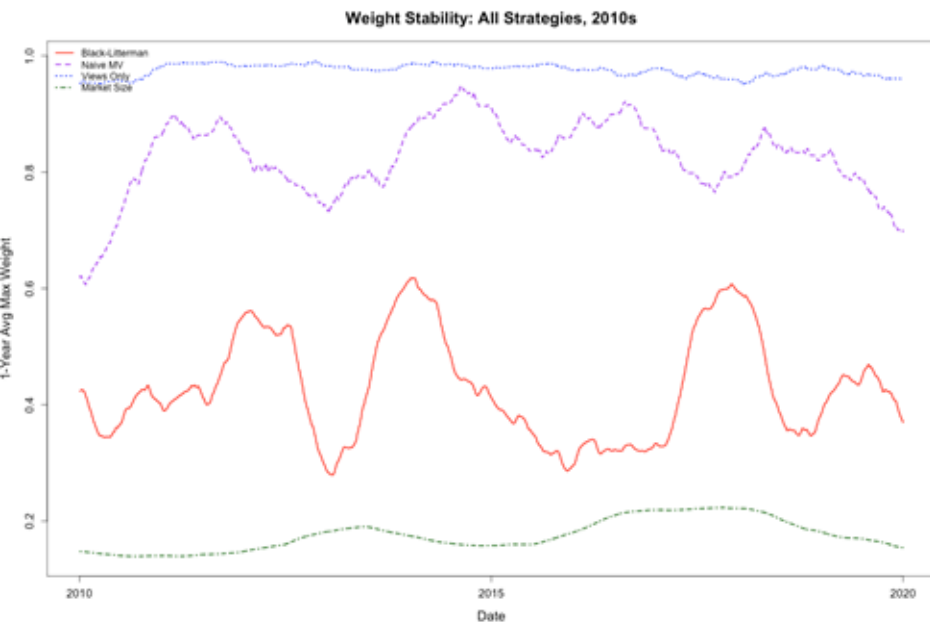
Trailing Weight Stability



Trailing Weight Stability



Trailing Weight Stability



Full Performance Comparison

Strategy	Ann. Return	Ann. Vol	Sharpe	Turnover	Concentration	Avg. Max Weight
Black-Litterman	0.3205	0.2650	1.2094	0.1117	0.4012	0.5058
Naive MV	0.6466	0.3701	1.7471	0.1296	0.7642	0.8200
Views Only	0.6245	0.3095	2.0179	0.1436	0.9643	0.9730
Market Size	0.3053	0.2177	1.4024	0.0009	0.0534	0.1499

Performance After 10 bps Transaction Cost

Strategy	Ann. Return	Ann. Vol	Sharpe	Turnover	Concentration	Avg. Max Weight
Black-Litterman	0.2655	0.2651	1.0015	0.1117	0.4012	0.5058
Naive MV	0.5827	0.3701	1.5745	0.1296	0.7642	0.8200
Views Only	0.5538	0.3094	1.7897	0.1436	0.9643	0.9730
Market Size	0.3049	0.2177	1.4003	0.0009	0.0534	0.1499

CAPM Attribution: BL vs MV

Strategy	Alpha Daily	Beta	Residual Std. Error	R ²
BL	0.000267	0.708836	0.009269	0.3973
Naive MV	0.000905	0.786743	0.014430	0.2509

CAPM Attribution: Diagnostic Strategies

Strategy	Alpha Daily	Beta
Views Only	0.000830	0.888078
Market Size	0.000194	0.849001

BL Top 10 Weights

Rank	Industry	BL Avg Weight	MV Avg Weight
1	Util	0.144500	0.026324
2	Smoke	0.114520	0.073747
3	Beer	0.077413	0.015346
4	Hardw	0.055538	0.011767
5	Gold	0.047901	0.125978
6	Steel	0.045994	0.013356
7	Softw	0.041430	0.039629
8	Food	0.039015	0.013545
9	Chips	0.031010	0.020904
10	RIEst	0.029473	0.018715

MV Top 10 Weights

Rank	Industry	BL Avg Weight	MV Avg Weight
1	Gold	0.047901	0.125978
2	Coal	0.015468	0.097445
3	Smoke	0.114520	0.073747
4	Soda	0.017160	0.055926
5	HIth	0.020670	0.043847
6	Mines	0.017419	0.041309
7	Oil	0.020390	0.040144
8	Softw	0.041430	0.039629
9	Guns	0.007997	0.035238
10	Txtls	0.003216	0.029452

BL: Investor View Comparison

BL View	Ann. Return	Ann. Vol	Sharpe	Turnover	Concentration	Daily Alpha	Beta	Largest Avg Weight
Trailing	0.3205	0.2650	1.2094	0.1117	0.4012	.0267%	0.7088	0.1445
Mean Reversion	0.3114	0.2650	1.1752	0.1118	0.4000	.0248%	0.7096	0.1436
Momentum	0.3116	0.2650	1.1761	0.1118	0.4001	.0248%	0.7095	0.1436